

VAMSI KRISHNA PASYAVULA

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PROFESSIONAL SUMMARY

GenAI Architect / Technical Architect with 20+ years designing enterprise, cloud-native, API-led and event-driven architectures for high-volume Telecom BSS/OSS platforms — Billing, Charging, Order Management, Product Catalog, Mediation, and Orchestration — combined with hands-on experience architecting GenAI and agentic AI solutions in production: LLM integration, Retrieval-Augmented Generation (RAG), tool/function calling, multi-agent orchestration (LangGraph), and Human-in-the-Loop guardrails. Skilled at evaluating when to apply prompting, RAG, tool calling, agents, or fine-tuning to a given use case, and at defining the security, privacy, and governance controls that keep GenAI solutions safe, scalable, and compliant. Delivers on Azure, AWS, and GCP with Kubernetes, Docker, and CI/CD (Terraform, Ansible, GitHub Actions), backed by strong observability (ELK Stack, Dynatrace) and IAM/RBAC security practices. Partners across product, engineering, data, security, and platform teams to turn ambiguous, open-ended problems into scalable, reusable, production-ready GenAI reference architectures.

Key Match: GenAI/Agentic AI Architecture (LLMs, RAG, Tool/Function Calling, LangGraph Multi-Agent Orchestration) • BSS/OSS/Orchestration Domain Expertise (Billing, Charging, Order Management, Product Catalog) • GenAI Guardrails, Security, Privacy & Governance (IAM/RBAC, HITL) • Enterprise & Distributed System Architecture • Kubernetes/Docker & CI/CD (Terraform, Ansible) • APIs & Event-Driven Integration (Kafka, RabbitMQ, REST) • Observability & Production Reliability (ELK, Dynatrace) • Cross-Functional Enterprise Collaboration

WORK EXPERIENCE

AT&T | Seattle, USA (Remote)

Apr 2025 - Present

Lead System Engineer / Technical Architect | Client: AT&T, USA

- Hands-on experience leveraging Generative AI and LLMs (GPT-4, Claude, GitHub Copilot) to automate and accelerate architecture design and development workflows.
- Strong architecture expertise in agentic AI platforms: multi-agent orchestration frameworks (LangGraph), tool/function calling, Retrieval-Augmented Generation (RAG), and Human-in-the-Loop (HITL) guardrails for enterprise-grade GenAI workflows.
- Architected Prompt Engineering pipelines, vector databases (Pinecone, pgvector), and semantic search at enterprise scale — deciding when prompting, RAG, or agents best fit a given use case.
- Apply identity and access management controls — AWS IAM and Azure Active Directory (Entra ID) with role-based access control (RBAC) — to define security, privacy, and governance guardrails for GenAI and enterprise platforms.
- Designed and delivered scalable technical architectures for Customer Account Management and Consent and Privacy platforms, ensuring alignment with business goals and system performance requirements.
- Involved in architectural transformation initiative for Omni-channel integration, enabling seamless convergence of Wireless and Wireline services.
- Delivered technical designs for applications like Commerce, Order Management, Billing and Payment Tools.
- Leading solution design for microservices and event-driven integrations across critical domains such as Web Account Management, Product and Offering, Customer Graph, Order Graph, and Amdocs Billing systems within complex eCommerce buy flows.
- Implement observability and performance engineering practices (ELK Stack, Dynatrace) across microservices and GenAI-enabled workflows for proactive bottleneck analysis, root-cause investigation, and production reliability.
- Contributed to automation of UML diagram generation by reverse-engineering microservices from source code using Sparx Enterprise Architect, improving technical documentation and system understanding.
- Participating in architectural reviews to present architecture slide decks for EPICs and provide technical solutions to stakeholders — design reviewers and business leaders.

Ericsson Inc. | Canada

Aug 2021 - Apr 2025

Senior Solution Architect & E2E Architect | Client: T-Mobile, USA

- Led end-to-end architecture and technical delivery of large-scale OSS/BSS transformation programs covering Billing, Charging, Order Management, Product Catalog, Mediation, and Provisioning systems.
- Architected and delivered scalable, cloud-native microservices on AWS using Spring Boot, Kubernetes, Docker, and Helm, ensuring high availability and performance.
- Designed and developed RESTful APIs and event-driven integration layers leveraging Kafka, RabbitMQ, Azure Event Hub, and PostgreSQL to enable secure, real-time data exchange across systems.
- Migrated legacy OSS components to modern microservices using event-driven architectures powered by Apache Kafka, and refactored performance-critical services to improve throughput and reduce latency.
- Integrated rule-based automation using AI-enabled configuration engines to optimize catalog management and order orchestration, reducing manual interventions by 35%.
- Proposed and developed PoC for AI-driven catalog recommendation engine, enabling intelligent bundling and guided selling aligned with user behavior analytics.
- Participated in solutioning discussions for predictive failure analytics in Order Care systems using operational event telemetry and ML models.
- Collaborated with stakeholders on architecture governance, risk mitigation, and milestone reviews.
- Served as the final authority on technical deliverables, ensuring cross-domain alignment, architectural compliance, and quality standards across Agile sprints.

- Mentored team members and promoted best practices in microservices, containerization and secure coding.
- Created and delivered Statements of Work (SOWs) and effort estimates for complex, enterprise-scale solutions; supported pre-sales engagements and advised clients on architecture strategy.
- Led cross-functional Agile teams through sprint planning, daily stand-ups, retrospectives, and demos, driving effective collaboration and delivery execution.
- Delivered integrated telecom solutions with Ericsson platforms (EOC, ECM, CS, EMA, BSCS, SAPC) and optimized configurations for CPQ, Rate Plans, and Order Negotiation.
- Led implementation of CI/CD pipelines and Infrastructure as Code using GitHub Actions, Jenkins, Terraform, and Ansible.

Ericsson Inc.
|
Seattle, USA
May 2018 – Aug 2021

Solution Architect II
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Client: T-Mobile, USA

- Delivered architecture and development of cloud-native telecom applications using microservices and Kubernetes for EOC (Order Management) and ECM (Product Catalog).
- Designed and delivered scalable microservices for order orchestration and fulfillment, integrated with CS16 (Charging), BSCS (Billing), and EMA (Provisioning).
- Customized the EOC order decomposition engine, achieving a 70% reduction in order fallout.
- Built and deployed an AI-assisted configuration engine PoC for automated catalog validation and dependency conflict resolution.
- Conducted impact analysis for major change requests using service contract versioning and regression techniques to maintain compatibility.
- Implemented event-driven transaction processing across OSS/BSS layers using Apache Kafka to improve system decoupling.
- Developed CI/CD pipelines leveraging Jenkins, Docker, Helm, and GitLab for reliable and frequent deployments.
- Collaborated with cross-domain teams and client stakeholders to ensure alignment of technical solutions with business goals.
- Supported L3/T3 maintenance during the Prepaid rollout, providing architectural oversight and resolving escalated issues.

Ericsson India Global Services Pvt Ltd
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Bangalore
Apr 2014 – May 2018

Solution Architect
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Client: T-Mobile USA, Ooredoo-Kuwait, DST-Brunei, MTN-Zambia, Banglalink-Bangladesh

Tech Stack: Java [8, 11, 17], Spring 5.x, Spring Boot 2.x, Oracle DB, DevOps CI/CD, REST APIs

- Led development of Postpaid foundation for OM (EOC) Module with new design for Order Decomposition and Order Orchestration.
- Involved in Order Management customization for Prepaid Solution; coordinated major CR impact analysis and delivery across internal and external domain teams.
- Developed critical components including Spring forms, controllers, JSP views, Hibernate entities, Spring-Hibernate DAO and Spring models following MVC architecture.
- Used Hibernate persistence framework on Oracle DB for billing details PDF generation; written extensive JUnit tests for comprehensive coverage.
- Exposed User Profile services to other departments using RESTful Web Services.
- Involved in Oracle Database design; developed APIs for integration with external systems.

EARLIER EXPERIENCE

Company / Period / Role	Tech Environment	Key Contributions
Enhancesys Innovations Pvt Ltd, Bangalore May 2012 – Apr 2014 Team Lead Airtel, India	J2EE, JBoss, Hibernate, Oracle 10g, Hadoop, HornetQ, Elasticsearch	• TMForum SID APIs, Product Catalog integration, JBoss clustering, CI/CD, UAT
TecnoTree Convergance Ltd, Bangalore Jan 2010 – May 2012 Team Lead Emtel (Mauritius), Claro (Peru, Central America)	J2EE, JBoss, Hibernate, Oracle 10g PL/SQL, PERL, Linux	• RTCC design, 3GPP/Diameter, Replenishment modules, production deployment
CSG International (fmr. INTEC Telecom), Bangalore Dec 2006 – Nov 2009 Development Consultant, Senior POLKOMTEL (Plus), Poland	Singl.eView, EPM, Core Java, PERL, Oracle, UNIX, SOAP	• Tariff/DA/EPM config, API & reporting framework, release management, production support
Lifetree Convergence Ltd, Bangalore Jul 2005 – Dec 2006 Senior Software Engineer Emtel, Claro, MTN-Namibia, Madacom	J2EE, Oracle 10g PL/SQL, Pro*C	• Pricing module, Billing/Mediation, rate plan migration, CDR revenue sharing

Company / Period / Role	Tech Environment	Key Contributions
BPL Telecom Ltd, Bangalore Aug 2003 – Jul 2005 Software Engineer BSNL, India	Pro*C, Oracle PL/SQL, UNIX	ICB + Fraud Management, BSNL South/North implementation, CRM backend

TECHNICAL SKILLS

GenAI & Agentic AI	LLM Integration (GPT-4, Claude, Gemini-family), Retrieval-Augmented Generation (RAG), Vector Databases (Pinecone, pgvector), Tool/Function Calling, Multi-Agent Orchestration (LangGraph), Prompt Engineering Pipelines, Human-in-the-Loop (HITL) Guardrails, AI-Assisted Development (Windsurf, GitHub Copilot)
BSS/OSS/Orchestration Domain	Billing, Charging, Order Management, Product Catalog, Mediation, Provisioning, CPQ & Order Orchestration — Ericsson (EOC, ECM, CS, EMA, BSCS, SAPC), Amdocs Billing
Architecture	Cloud-Native Design, Microservices, Event-Driven Architecture, Serverless, API Management, Architecture Modeling (C4, UML, Sequence & Data Flow Diagrams, Domain Models via Sparx EA), HLD/LLD, Reference Architectures
Cloud & Platforms	Azure, GCP, AWS, Kubernetes, Docker, Hybrid & Multi-Cloud Architecture
AI / ML	Python, Scikit-learn (PoC), TensorFlow (Conceptual)
Integration & Messaging	Kafka, RabbitMQ, Azure Event Hub, REST APIs, Event-Driven Integration Patterns
DevOps & Automation	CI/CD, GitHub Actions, Jenkins, Terraform, Ansible, CloudFormation, BPMN Tools
Observability & Monitoring	ELK Stack (Elasticsearch, Logstash, Kibana), Dynatrace, Performance Engineering, Capacity Planning
Security & Identity	AWS IAM, Azure Active Directory (Entra ID), RBAC, GenAI Guardrails, Secure Coding Practices
Programming & Frameworks	Java, Python, Spring Boot, Hibernate, Pro*C, Camunda
Database	Oracle (PL/SQL), PostgreSQL, Hadoop, MongoDB
Standards & Practices	Agile/SAFe, TMForum SID/TAM, TOGAF, Sparx Enterprise Architect

CERTIFICATIONS

Architecting Google Cloud Platform | Azure AI Fundamentals | AWS Certified Solutions Architect – Associate | L2 AI/ML Assessment (Ericsson Academy) | SAFe Agile – Product Owner & Scrum Master | Ericsson Technical Certified Professional | IBM Big Data & Hadoop | Cloud-U Rackspace

EDUCATION

- M.Sc. Computer Science — 2003 — Sri Krishnadevaraya University
- B.Sc. Computer Science — 2001 — Sri Krishnadevaraya University

Canadian Permanent Resident — fully authorized to work anywhere in Canada, no employer sponsorship required. Open to relocation: Montreal, QC.